



CMedRAGBot: A Chinese Medical Chatbot Based on Graph RAG and Large Language Models

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Abstract

In the domain of Chinese clinical medical question-answering (QA), traditional Large Language Models (LLMs) encounter challenges such as hallucinations and difficulties in updating knowledge for knowledge-intensive tasks. To address these issues, this research presents a Chinese clinical medical QA model that integrates Retrieval-Augmented Generation (RAG) and a medical knowledge graph, named CMedRAGBot. First, a Chinese medical knowledge graph encompassing multiple entity types—including diseases, medications, and symptoms—is constructed. Based on this knowledge graph, a Named Entity Recognition (NER) model built on a Chinese-RoBERTa and BiGRU architecture is designed, with data augmentation strategies employed to enhance its generalization capability. In addition, prompt engineering techniques are used to implement intent recognition for user queries, mapping them to predefined intent categories. Finally, the aforementioned modules are integrated to form a complete Chinese clinical medical QA system. In the experimental evaluation, CMedRAGBot is deployed on five state-of-the-art LLMs (including ChatGPT-4o, ChatGPT-o1, DeepSeek-R1, Llama-3.3-70B-Instruct, and Gemini 2.0 Flash) and tested using specialized question banks derived from the Chinese Clinical Medical Qualification Examinations and Residency Standardization Training Examinations from 2000 to 2023. The results indicate that the integration of CMedRAGBot significantly improves the test accuracy of all models, with increases of up to approximately 10%. Furthermore, ablation experiments reveal that data augmentation enhances NER model's F1 score from 95.27% to 97.55%, while the inclusion of an intent recognition module markedly improves the model's ability to understand complex queries, thereby further boosting answer accuracy. Source code of the research is available at <https://github.com/zhdongfang/CMedRAGBot>.

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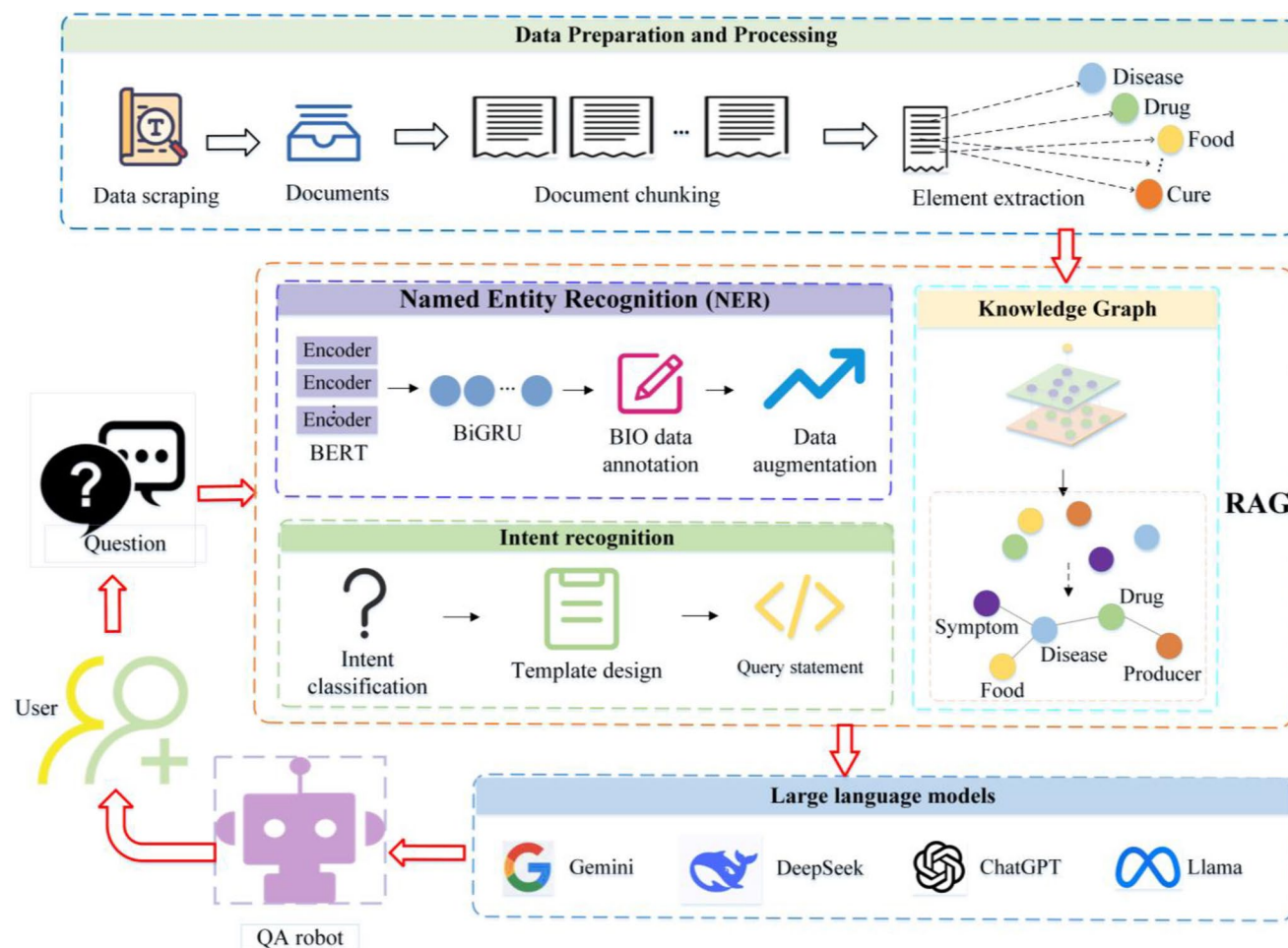
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Graphical Abstract

The proposed CMedRAGBot mainly includes four steps: (1) A Chinese medical knowledge graph is constructed to encompass multiple entity types such as diseases, medications, and symptoms. (2) An NER model based on Chinese-RoBERTa and BiGRU is designed, with data augmentation to enhance generalization capability. (3) Prompt engineering is employed to identify user intent and map queries to predefined intent categories. (4) The aforementioned modules are integrated with LLMs to build CMedRAGBot. This method effectively addresses the knowledge acquisition, comprehension, and hallucination challenges faced by traditional LLMs in Chinese medical question answering, thus providing novel insights for intelligent diagnosis and treatment



Keywords Retrieval-augmented generation · Chinese medical chatbot · Large language model · Knowledge graph · Named entity recognition

1 Introduction

In recent years, Large Language Models (LLMs) have attracted significant attention for their potential applications in the medical field. By analyzing and processing large volumes of data, these models offer effective decision support to medical professionals, thereby enhancing the quality and efficiency of clinical services [1–3]. However, several challenges hinder their practical implementation, with the most prominent being the problem of ‘hallucinations’ [4,

5]. Because LLMs generate text probabilistically, their content is based on statistical correlations rather than verified factual information. Consequently, these models frequently produce outputs that, although appearing logical, may lack factual accuracy. Furthermore, when supporting data is insufficient, these models may speculate or fill in gaps, potentially resulting in inaccurate or false information, which can mislead clinical decisions and pose risks to patients [6, 7].

In the field of medical question-answering (QA), Retrieval-Augmented Generation (RAG) technology has demonstrated significant potential for improving information access and response accuracy. RAG integrates information retrieval with natural language generation to overcome the limitations of pre-trained LLMs in knowledge-intensive tasks [8–11]. Traditional RAG models incorporate external knowledge bases to retrieve relevant information and generate high-quality responses to user queries. Lewis et al. [12] initially proposed RAG as a solution to the limitations of large models in accessing and manipulating external knowledge. The original RAG focuses on retrieving answers from specific text regions but faces challenges when handling complex graph structures, leading to underperformance in aggregating semantic knowledge from diverse data sources. Since then, numerous researchers have explored combining RAG with medical knowledge bases to enhance its applicability in the medical domain. Xiong et al. [13] proposed a model named MIRAGE, which demonstrated that combining different medical corpora and retrieval systems could enhance the accuracy of LLMs using MedRAG toolkit. Ye [14] introduced a novel approach involving learned ranking mechanisms to improve the performance of RAG-based electronic medical record search engines. Zhou et al. [15] employed RAG technology by fine-tuning an embedding model on a corpus of 25 gastrointestinal disease guidelines and integrating it with the llama-index to develop the gastroenterology QA system.

Traditional RAG models commonly utilize vector databases. However, in the clinical medical domain, queries frequently involve complex contexts and multi-layered relationships that vector similarity measures may fail to capture accurately. This limitation is particularly critical in healthcare, where patient needs and intents are often intricate and context-dependent [16, 17]. Furthermore, the effectiveness of vector databases is heavily dependent on the quality and scope of the training data. In the rapidly evolving medical field, inadequate representation of the latest medical advancements in training data can lead to inaccurate or irrelevant retrieval outcomes [18, 19]. Moreover, existing research has shown that LLMs demonstrate strong multilingual capabilities, often performing at comparable levels across different languages in medical examinations [20]. Nevertheless, our work is motivated by the need for more precise, application-oriented solutions specific to Chinese clinical medicine. Accordingly, we focus on scenarios involving Chinese medical QA. Chinese medical texts have distinctive linguistic structures and knowledge systems, yet current open-source corpora and medical knowledge graphs predominantly cater to English. As a result, there is a relative shortage of structured Chinese medical data, making it difficult for multilingual LLMs to fully grasp and generate high-quality responses in Chinese.

To address these challenges, we extend the work of Hu et al. [21] by integrating RAG with knowledge graphs, proposing a clinical medical QA framework that leverages RAG and LLMs. The goal is to improve the accuracy and reliability of Chinese medical QA systems. We first construct a knowledge graph for clinical diseases, encompassing multi-dimensional information, including symptoms, causes, and treatment plans. Next, we employ BERT model to enhance the entity recognition process, thereby improving the model's comprehension of medical text. For the entity recognition task, we propose and implement three data augmentation strategies: entity substitution, entity masking, and entity concatenation. These strategies significantly improve RoBERTa model's performance in entity recognition, enhancing its robustness for real-world applications. Additionally, we employ prompt engineering techniques to categorize user intent into predefined categories, designing tailored prompts for each. This approach enables the model to analyze user intent more effectively, resulting in more accurate responses. By incorporating contextual learning and chain-of-thought techniques, we achieve significant improvements in intent recognition accuracy and the logical consistency of generated answers, ultimately providing effective solutions for clinical medical QA.

This research aims to offer new perspectives and methodologies for applying LLMs in healthcare, thereby establishing a solid theoretical and practical foundation for future studies.

2 Methods

This work presents a Chinese medical QA model that integrates RAG and LLMs, leveraging existing Chinese medical datasets to construct a knowledge graph as the structured retrieval foundation for augmentation. Upon receiving a Chinese-language query, the model performs sequential entity recognition and intent classification, subsequently generating contextual graph queries against the knowledge graph and structured prompts incorporating retrieved medical evidence, thereby enhancing LLM's capability to comprehend and respond to clinically complex inquiries. The architectural overview is illustrated in Fig. 1.

2.1 Data Preparation and Processing

To utilize existing Chinese medical datasets, this research adopts the methodology from OpenKG (<https://github.com/nuolae/disease-kb>) for data acquisition. In October 2024, we performed dynamic web crawling and systematic data collection from “XunyiWenyao” medical platform. The

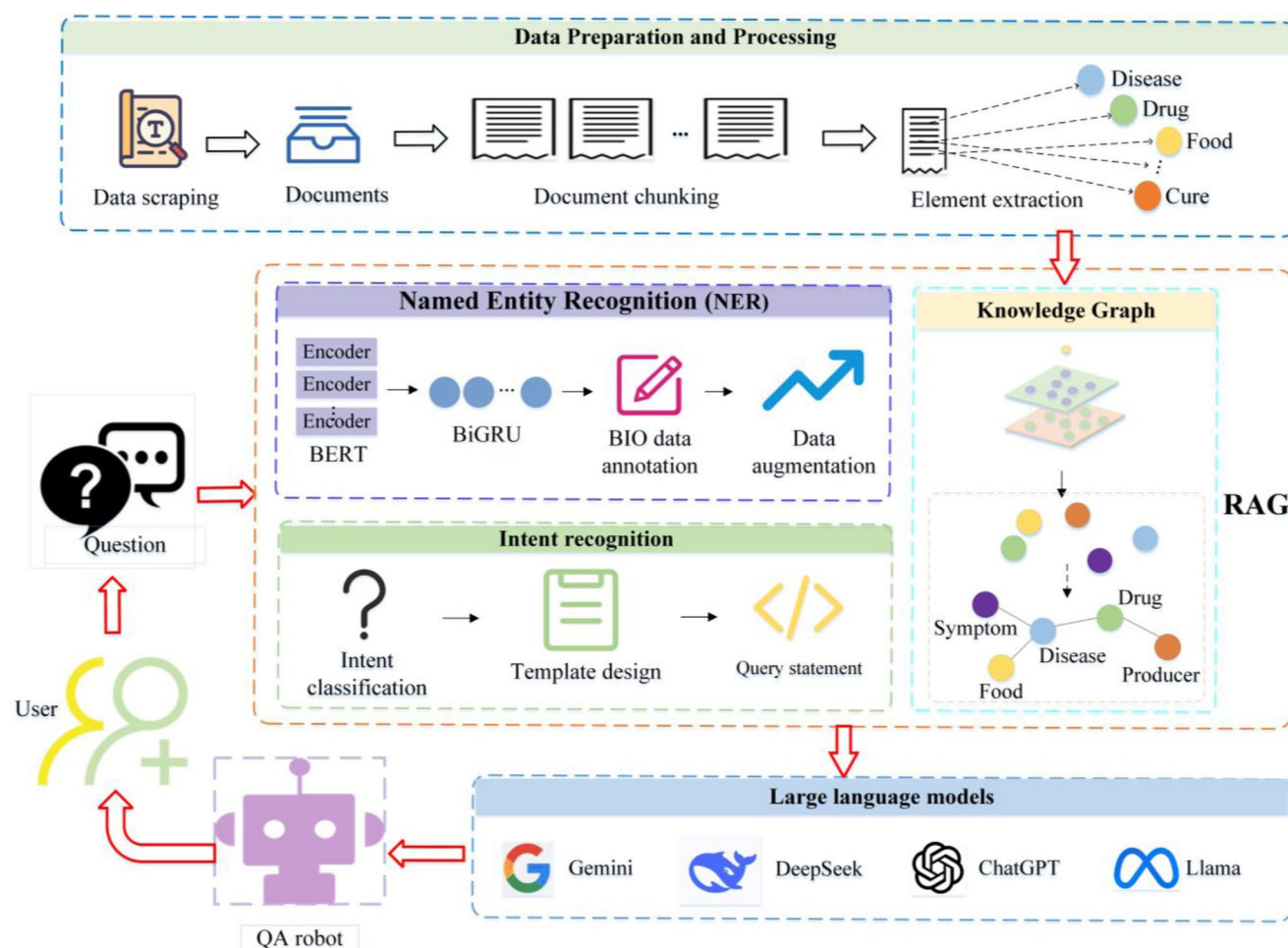


Fig. 1 CMedRAGBot framework

collected data underwent rigorous cleansing, transformation, and structural organization to meet knowledge graph requirements, resulting in a schema encompassing eight entity types, including diseases, symptoms, drugs, and diagnostic procedures. To enhance semantic granularity, we implement a hybrid text segmentation strategy combining maximum forward matching, reverse matching, and bidirectional maximum matching algorithms, supplemented by heuristic rules for resolving ambiguous boundaries. This approach ensures coherent semantic chunks with minimized contextual fragmentation.

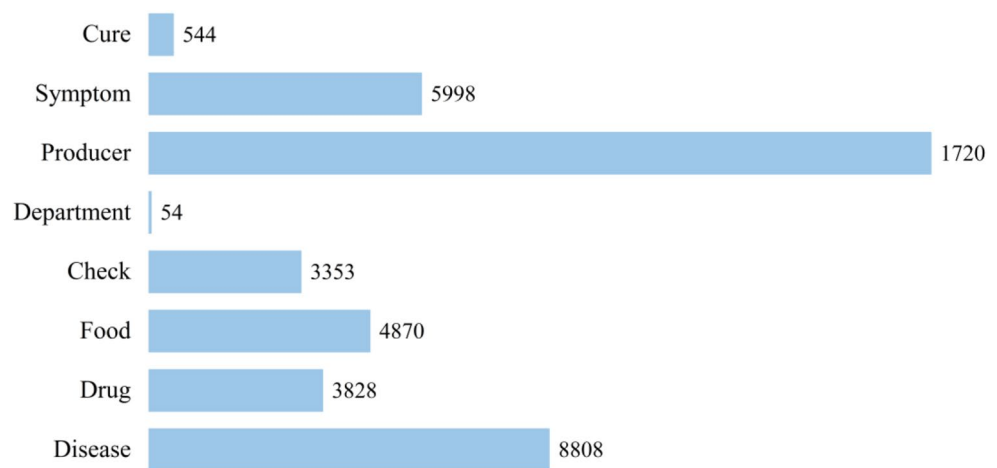
Furthermore, based on the constructed lightweight knowledge graph, CPubMed-Wiki API (<https://cpubmed.openi.org.cn/graph/wiki>) is also invoked. The knowledge content provided by this interface originates from CPubMed-KG graphs, which is collaboratively constructed by five institutions, including Peking Union Medical College Hospital, and is used to supplement the lightweight knowledge graph. The knowledge in this graph was most recently updated on February 27, 2025.

2.2 Knowledge Graph

2.2.1 Element Extraction

Element extraction is a critical step in constructing the knowledge graph, involving the identification and extraction of meaningful entities and their attributes from raw data [22]. The preprocessed structured medical data is stored in JSON format, where each entry represents a record containing a disease and its related information. For each entity type, the corresponding fields are extracted. During extraction, it is crucial to handle data anomalies and irregularities, such as nested lists, removing ellipses from symptom names, and correcting data format errors. In the element extraction process, we primarily extract eight entity types from the dataset, as summarized in Fig. 2. In constructing the knowledge graph, despite significant imbalances in the distribution of entity categories, we achieve effective results.

For disease entities, we extract multiple key attributes to enrich node information within the knowledge graph, as detailed in Table 1.

Fig. 2 Entities in the knowledge graph**Table 1** Disease entity attributes

Entity	Description
Name	Represents the name of a disease
Description	Provides a brief description of the disease, offering key information such as what it is and how it affects the body
Cause	Refers to the causes of the disease, includes the underlying factors or conditions that lead to the disease, such as infections, genetic mutations, or environmental influences
Prevention	Describes strategies or actions that can be taken to reduce the risk of developing the disease or prevent its spread
Treatment duration	Provides information on the estimated length of time needed for treatment, Indicates how long a patient may need to undergo therapy or medical intervention
Cure rate	Represents the probability of recovery, usually expressed as a percentage
Susceptible populations	Refers to susceptible populations, the groups of people who are more likely to contract the disease due to various factors such as age, genetics, or lifestyle

2.2.2 Hierarchy Linking

Hierarchical linking establishes relationships between entity categories, clarifying classification and affiliations to enhance the organizational and semantic representation capabilities of the knowledge graph [23, 24]. Extracting eight distinct entity categories allows us to establish a basic hierarchical structure. In this study, we utilize both entity attribute information and inter-entity relationships to further develop hierarchical links. The primary hierarchical links in our knowledge graph are as follows.

- Hierarchical relationship between diseases and medical subjects: This link connects disease entities with their respective medical subjects, establishing a hierarchy that relates diseases to subjects. For instance,

linking the disease ‘hypertension’ to the subject ‘cardiology’ indicates that hypertension falls within the field of cardiology.

- Hierarchical relationship between symptoms and diseases: Symptoms, as manifestations of diseases, are linked to their corresponding diseases to create a hierarchical structure connecting diseases to symptoms.

2.2.3 Relationship Linking

Relationship linking is fundamental to the knowledge graph, aiming to define associations between entities and enrich the graph’s semantic network. The relationship linking process involves relationship extraction and representation, data deduplication, and anomaly handling. In the knowledge graph, relationships are represented as edges connecting two entity nodes, each carrying a specific relationship label [25–27]. Initially, we traverse data records to identify relationships between entities based on field content. For instance, relationships between diseases and drugs are extracted from the ‘common_drug’ and ‘recommended_drug’ fields. When creating disease nodes, attributes are stored as key-value pairs within each node. For entities such as drugs, food, and symptoms, only their names are initially stored, with other attributes added as needed.

To improve efficiency, we use batch processing for relationship creation, dynamically generating Cypher queries via Python string formatting to create nodes and relationships based on entity and relationship information. During relationship extraction, data structures such as sets are employed to remove duplicate relationships, and special cases or data anomalies are addressed to ensure the completeness of the relationship links, including handling irregularities in symptom names or missing drug manufacturer information.”

Additionally, optimization measures for indexing and resource management are implemented during the

Table 2 Relationships in the knowledge graph

Relationship	Count	Description
Association	8,843	The association of a disease with a medical department. Example: Internal Medicine, associated with, Respiratory Internal Medicine
Common_drug	14,647	The commonly used drugs for a disease. Example: Acute Respiratory Distress Syndrome, commonly used, Human Albumin
Recommended_food	22,230	Foods that are recommended for consumption in relation to a disease. Example: Acute Respiratory Distress Syndrome, recommended food, Lotus Seeds
Available_drug	17,315	Drugs that are available for sale. Example: Human Albumin, available for sale, Lask Human Albumin
Need_check	39,418	The diagnostic tests required for a disease. Example: Unilateral Pulmonary Emphysema, required check, Bronchography
prohibited_food	22,239	Foods that should be avoided in relation to a disease. Example: Unilateral Pulmonary Emphysema, required check, Bronchography
recommended_drug	59,465	Drugs that are recommended for the treatment of a disease. Example: Hemorrhoids, recommended drug, Jingwanhong Hemorrhoid Cream
Recommended_recipe	40,221	Recommended dietary recipes for a disease. Example: Acute Respiratory Distress Syndrome, recommended recipe, Lily Sugar Porridge
Has_symptom	54,710	The symptoms and signs associated with a disease. Example: Acute Respiratory Distress Syndrome, symptom, Difficulty Breathing
Comorbidity_with	12,024	Diseases that often occur as comorbidities. Example: Acute Respiratory Distress Syndrome, comorbidity, Bacterial Pneumonia
Treatment_method	21,047	The treatment methods used for a disease. Example: Acute Lung Abscess, treatment method, Antibiotic Treatment

relationship linking process. Indexes are created in Neo4j for frequently queried fields, such as entity names, to enhance the system's efficiency in locating relationships.

Batch sizes are configured for bulk creation of nodes and relationships to expedite data processing. From the dataset, key relationships are extracted and established, as summarized in Table 2.

2.3 Named Entity Recognition (NER)

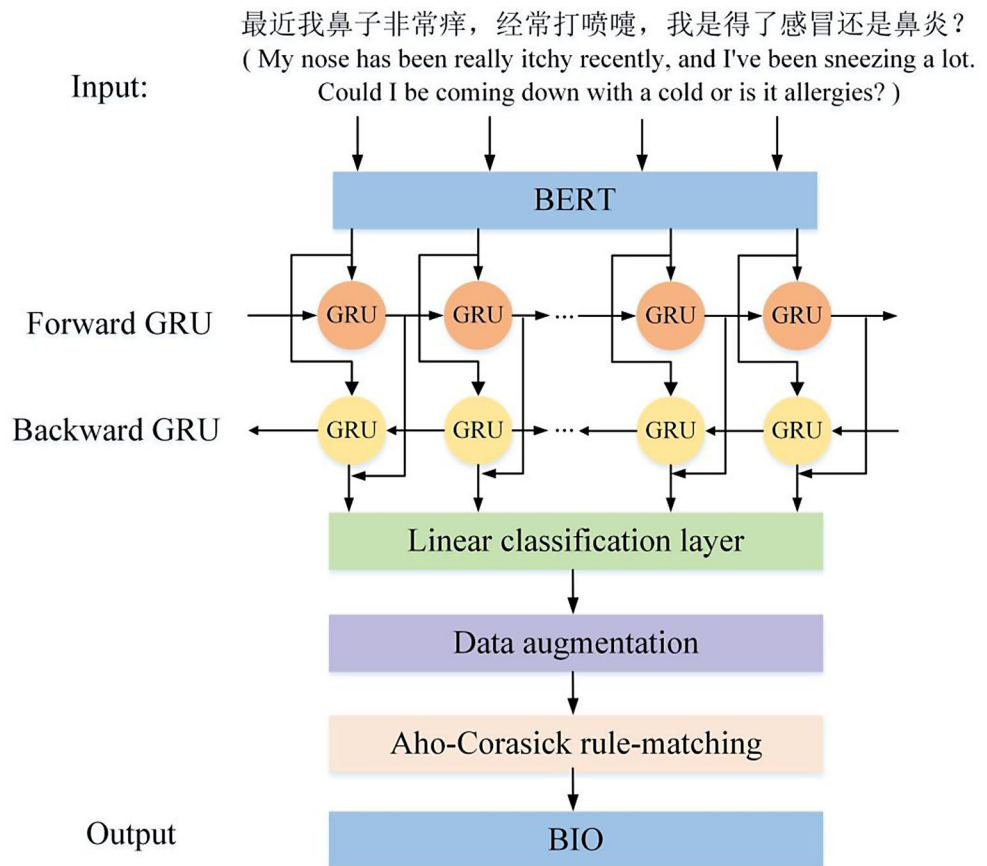
In the proposed model, NER module extracts key medical entities from user queries such as “hypertension” “head-ache” “antibiotics”, providing structured semantic anchors for subsequent retrieval and generation steps. The accuracy of NER component is critical to the semantic comprehension capabilities of RAG system. In complex medical domains, overlooking crucial information such as diseases or medications can lead to reasoning biases within LLMs, resulting in potentially misleading answers due to insufficient context [28, 29]. To address this issue, the study integrates NER module with a knowledge graph at a deep level. For instance, when a user asks “What breakfast foods should patients with hypertension avoid?”, NER module identifies “hypertension” as the key entity and triggers the knowledge graph to retrieve relevant attributes—such as restricted foods or drug interactions—associated with that entity. LLMs then synthesize these multi-dimensional contextual details to produce logically coherent responses.

We employ BERT model for entity recognition and incorporate data augmentation and rule-matching modules to enhance its performance and generalization capability. BERT model captures bidirectional contextual semantic information from input sentences using Transformer architecture and attention mechanisms to effectively extract high-quality feature representations [30–32]. A bidirectional GRU (BiGRU) further processes BERT's output, modeling long-distance dependencies within the sequence to ensure seamless transmission and integration of semantic information.

In the final stage of NER model, a linear classification layer assigns entity labels to each word. Additionally, data augmentation strategies, including entity replacement, masking, and concatenation, are applied to expand the training dataset. The Aho-Corasick rule-matching module is integrated with the model's output to achieve rule-based entity supplementation and validation, as illustrated in Fig. 3.

2.3.1 NER Dataset

An NER dataset is constructed and annotated using BIO scheme. Rule-based matching techniques are integrated to precisely identify medical entities, including diseases, symptoms, diagnostic items, and medications. Specifically, Aho-Corasick automaton is employed for pattern matching against predefined entity dictionaries, thereby facilitating the annotation of entities within the text. The entity type

Fig. 3 NER framework

set is defined as $\mathcal{C} = \{c_1, c_2, \dots, c_n\}$. For each $c_j \in \mathcal{C}$, a corresponding entity dictionary $D_c = \{\omega_{c,1}, \omega_{c,2}, \dots, \omega_{c,n_c}\}$ is created. These dictionary entries are inserted into Aho-Corasick automaton to build a multi-pattern matching structure. When processing a text T of length n , the automaton outputs a set of matched entity intervals $\mathcal{M} = \{(s_j, e_j, c_j) \mid j = 1, \dots, m\}$, where s_j and e_j denote the start and end positions of an entity in T , and $c_j \in \mathcal{C}$ represents the entity type. Overlapping entities are resolved by prioritizing longer entities and sequentially annotating them to generate the final entity set.

Text paragraphs are extracted from multiple medical sources, including disease descriptions, preventive measures, and etiological explanations. Aho-Corasick matching process is applied to rapidly identify and annotate entities. To handle long texts effectively, we developed a punctuation-based segmentation method that randomly splits lengthy texts into shorter sentences, introducing variability to enhance dataset diversity. The resulting short text segments serve as inputs for NER annotation [33]. To address homonyms and ambiguities, contextual information is incorporated into the matching rules. For instance, polysemous terms are disambiguated based on surrounding words. Additionally, manual validation is conducted through random sampling checks to ensure the quality of automated annotations.

During NER annotation process, entities are labeled using BIO scheme:

- B-(Begin): Marks the starting position of an entity.
- I-(Inside): Marks positions within an entity.
- O-(Outside): Denotes non-entity tokens.

For a text $T = \{t_1, t_2, \dots, t_n\}$, the label sequence $L = (l_1, l_2, \dots, l_n)$ is initialized with $l_i = O$. If an entity of type c_j is matched in the interval $[s, e]$, the labels are updated as $l_s = B - c_j$ and $l_k = I - c_j, \forall k \in [s + 1, e]$. This ensures precise token-level entity tagging. For instance, in the sentence “The patient was diagnosed with chronic obstructive pulmonary disease and experiences frequent shortness of breath and chest tightness.”, the entity “chronic obstructive pulmonary disease” is labeled as a disease entity, “frequent shortness of breath” and “chest tightness” are labeled as symptom entities. The annotated result is presented as Table 3.

Finally, the annotated dataset is saved as an NER file for use in the subsequent training of the entity recognition model.

2.3.2 Data Augmentation

To enhance the generalization capability of the model, multiple data augmentation techniques are employed, including

Table 3 BIO result

Word	Tag
The	O
patient	O
was	O
diagnosed	O
with	O
chronic	B-Disease
obstructive	I-Disease
pulmonary	I-Disease
disease	I-Disease
and	O
experiences	O
frequent	B-Symptom
shortness	I-Symptom
of	I-Symptom
breath	I-Symptom
and	O
chest	B-Symptom
tightness	I-Symptom

entity replacement, entity masking, and entity concatenation. Pattern matching via automata is utilized to rapidly extract entity information from text, combined with TF-IDF-based entity alignment. Furthermore, through label analysis, entities in the text are automatically identified and extracted to prepare for subsequent training.

During data processing, the code first loads and segments text by randomly splitting long texts into shorter sentences, facilitating efficient model training. Concurrently, Aho-Corasick automaton is applied to recognize medical entities in the text and annotate them with B-entity type or I-entity type labels. To further improve semantic consistency, TF-IDF-based alignment is implemented to ensure high semantic similarity among identified entities. For each entity type $c_j \in \mathcal{C}$, an entity dictionary is defined as $D_c = \{\omega_{c,1}, \omega_{c,2}, \dots, \omega_{c,n_c}\}$, and the TF-IDF vector mapping for the vocabulary of type c_j , trained on the corpus, is formulated as Eq. (1).

$$\text{TF-IDF}_c : D_c \rightarrow \mathbb{R}^{(d)} \quad (1)$$

All entity terms are collectively fitted to generate $\{\mathbf{v}_{c,1}, \mathbf{v}_{c,2}, \dots, \mathbf{v}_{c,n_c}\}$, where $\mathbf{v}_{c,i} \in \mathbb{R}^{(d)}$. For a detected entity text “ent”, the corresponding TF-IDF model computes its vector $\mathbf{v}_{\text{ent}}$. The cosine similarity between $\mathbf{v}_{\text{ent}}$ and all entity vectors in the dictionary is calculated using Eq. (2).

$$\text{sim}(\mathbf{v}_{\text{ent}}, \mathbf{v}_{c,i}) = \frac{\langle \mathbf{v}_{\text{ent}}, \mathbf{v}_{c,i} \rangle}{\|\mathbf{v}_{\text{ent}}\| \|\mathbf{v}_{c,i}\|} \quad (2)$$

The maximum similarity score is denoted by $\text{max-score} = \max_i \text{sim}(\mathbf{v}_{\text{ent}}, \mathbf{v}_{c,i})$. If $\text{max-score} \geq 0.5$,

the entity text is aligned to the corresponding dictionary entry, thereby enhancing entity recognition precision.

To improve the generalization and robustness of NER model, three data augmentation strategies are defined:

- **Entity Replacement:** Randomly substitute entities in the original text with different entities of the same type [34].
- **Entity Masking:** Randomly delete or replace entity characters with special tokens (e.g., “[MASK]”) to simulate noisy data. For example, “患者出现头痛” (The patient experiences headache) is transformed into “患者出现[MASK]” (The patient experiences [MASK]). This strategy forces the model to rely more on contextual cues.
- **Entity Concatenation:** Augment text complexity by appending additional entities of the same type, connected via conjunctions such as “and”, “or”, or “as well as”. This helps the model learn to handle co-occurring entities.

2.3.3 NER Model Training

A pre-trained model-based NER architecture is employed, utilizing the Chinese pre-trained language model ‘Chinese-roberta-wwm-ext’. This model is combined with BiGRU and trained under a supervised learning framework with data augmentation. During inference, its weights remain unchanged.

Initially, the input Chinese sentences are tokenized, with special tokens [CLS] and [SEP] added to demarcate sequence boundaries. The tokenized sequence is mapped to word embeddings, denoted as $\mathbf{X} = [x_1, x_2, \dots, x_L]$, where L is the sequence length. These embeddings are fed into the encoder to obtain hidden representations $\mathbf{H} \in \mathbb{R}^{L \times d_{\text{BERT}}}$, with $d_{\text{BERT}} = 768$. The hidden vectors $\mathbf{H}$ are further processed by BiGRU to produce contextualized representations $\mathbf{R} \in \mathbb{R}^{L \times d_{\text{GRU}}}$, where d_{GRU} is GRU output dimension. Subsequently, the vector at each time step is mapped to the label space via a linear layer, yielding $P_i \in \mathbb{R}^K$, where K denotes the number of tags, $i = 1, 2, \dots, L$. Given the ground-truth label sequence $\hat{Y}^{(b)} \in \{0, 1, \dots, K-1\}^L$, the cross-entropy loss is computed as Eq. (3).

$$\mathcal{L} = \frac{1}{B \times L} \sum_{b=1}^B \sum_{i=1}^L \mathcal{L}_{\text{CE}}(P_i^{(b)}, \hat{Y}_i^{(b)}) \quad (3)$$

During training, $\mathcal{L}$ is backpropagated for gradient descent. At inference, predictions are derived as Eq. (4).

$$\hat{Y}_i^{(b)} = \arg \max_k P_{i,k}^{(b)} \quad (4)$$

Adam optimizer is employed with weight decay to mitigate overfitting. To ensure stability, especially on complex medical data, the initial learning rate is set to 0.00001, accompanied by a learning rate decay schedule. Training hyperparameters include a batch size of 60 and 30 epochs to guarantee stable convergence. Mixed-precision training is utilized to accelerate computations, while label smoothing reduces sensitivity to label noise by smoothing the ground-truth distributions [35, 36].

In summary, the model is initialized by loading pre-trained parameters from the *Chinese-roberta-wwm-ext* backbone, followed by supervised training on the augmented dataset. During training, validation performance is monitored in real time, with early stopping triggered if no improvement persists for consecutive epochs. Adam optimizer with weight decay is applied alongside a conservative initial parameters to balance convergence stability and medical data complexity. Mixed-precision training accelerates computation while label smoothing mitigates label noise sensitivity. After 30 epochs, the model is evaluated on the test set, where the confusion matrix analysis quantifies recognition disparities across entity categories. The complete training procedure is formalized in Algorithm 1.

Algorithm 1 NER training with pretrained encoder & BiGRU

Input: $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ (dataset), L (maximum sequence length), B (batch size)
 p (number of epochs), η (learning rate), E (pre-trained encoder),
 tag2idx (tag mapping)

For each (x_i, y_i) in $\mathcal{D}$ do:

$x'_i \leftarrow$ Tokenize (x_i) with [CLS] and [SEP], pad to length L or truncate to length L

$y'_i \leftarrow$ Map y_i to indices using tag2idx

Split $\mathcal{D}$ into $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{val}}$

Initialize model: $\theta \leftarrow \{E, \text{BiGRU}, f\}$

For epoch = 1 to p do:

Shuffle $\mathcal{D}_{\text{train}}$

For each batch (x_i, y_i) in $\mathcal{D}_{\text{train}}$ do:

$H \leftarrow E(x_i, \text{mask})$

$R \leftarrow \text{BiGRU}(H)$

$Z \leftarrow f(R)$

$\mathcal{L} \leftarrow \text{CrossEntropy}(Z, Y)$

Update θ using Adam optimizer

For each (x_i, y_i) in $\mathcal{D}_{\text{val}}$ do:

$y_{\text{pred}} \leftarrow \text{argmax}(f(\text{BiGRU}(E(x_i, \text{mask}))))$

Compute F_1 on $\mathcal{D}_{\text{val}}$

If $F_1 > F_{1, \text{best}}$ then:

$F_{1, \text{best}} \leftarrow F_1$

$\theta^* \leftarrow \theta$

Return θ^*

$\mathcal{D}$ represents the dataset of sentence-label pairs, where x_i is an input sentence and y_i is the corresponding label sequence. The symbols x'_i and y'_i denote the preprocessed versions of x_i and y_i , achieved by tokenization (including [CLS] and [SEP]) and padding or truncation to a fixed maximum sequence length L . E is the pre-trained encoder that produces hidden representations $H \in \mathbb{R}^{L \times d_{\text{BERT}}}$. The model parameters, denoted by θ , comprise the encoder E , a BiGRU whose output $R \in \mathbb{R}^{L \times 2h}$ captures sequential features, and a linear layer f that maps R to logits $Z \in \mathbb{R}^{L \times K}$, where K is the number of tags.

2.4 Intent Recognition

An intent recognition method based on LLMs is designed, integrating prompt engineering techniques to map user queries to predefined intent categories.

Initially, user query intents are classified into all categories, encompassing common needs and issues related to medical consultations. These categories include symptom inquiries, disease diagnosis, treatment plans, medication information, preventive measures, diagnostic tests, and rehabilitation advice, among others. Clearly defining intent categories allows the system to retrieve and generate appropriate responses more effectively.

When user inputs a question, the model employs prompts to guide the model in determining the intent category of the query. Contextual learning and chain-of-thought techniques are utilized to progressively analyze the underlying intent of the user's question, ensuring that the final output aligns with the user's needs [37–39]. As illustrated in Fig. 4, when a user asks, 'I have a cold, what should I do?', the system automatically identifies the user's intents, such as 'query disease overview', 'query treatment methods', and 'query necessary medications' among others.

To achieve intent recognition, various methods such as rule matching, model training, and prompt engineering from previous research are compared, leading to the adoption of a prompt engineering strategy [40–42]. Specific prompt templates are designed for each intent category, incorporating key information to guide LLMs in intent classification. For instance, for a symptom inquiry intent, a prompt might be 'Please determine whether the user's query involves a symptom description and classify it as a symptom inquiry intent.' The user's original query is combined with the prompt template and inputted into LLM, enabling the model to output the corresponding intent category based on prompt information and contextual content, as illustrated in Fig. 5.

Initially, a pre-trained BERT model is used to extract entities from the user's question. A comprehensive repository of medical knowledge is stored in an Neo4j database, and medical information related to these entities is

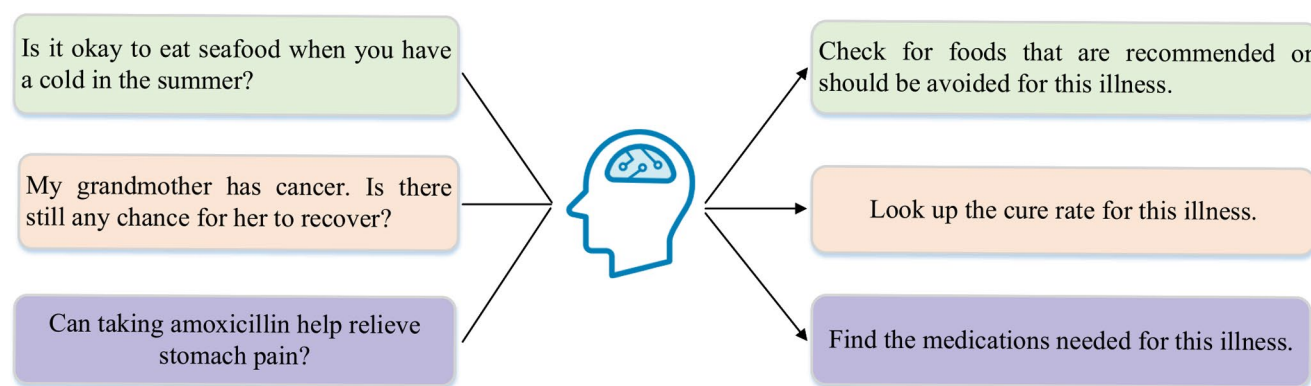


Fig. 4 Intent recognition

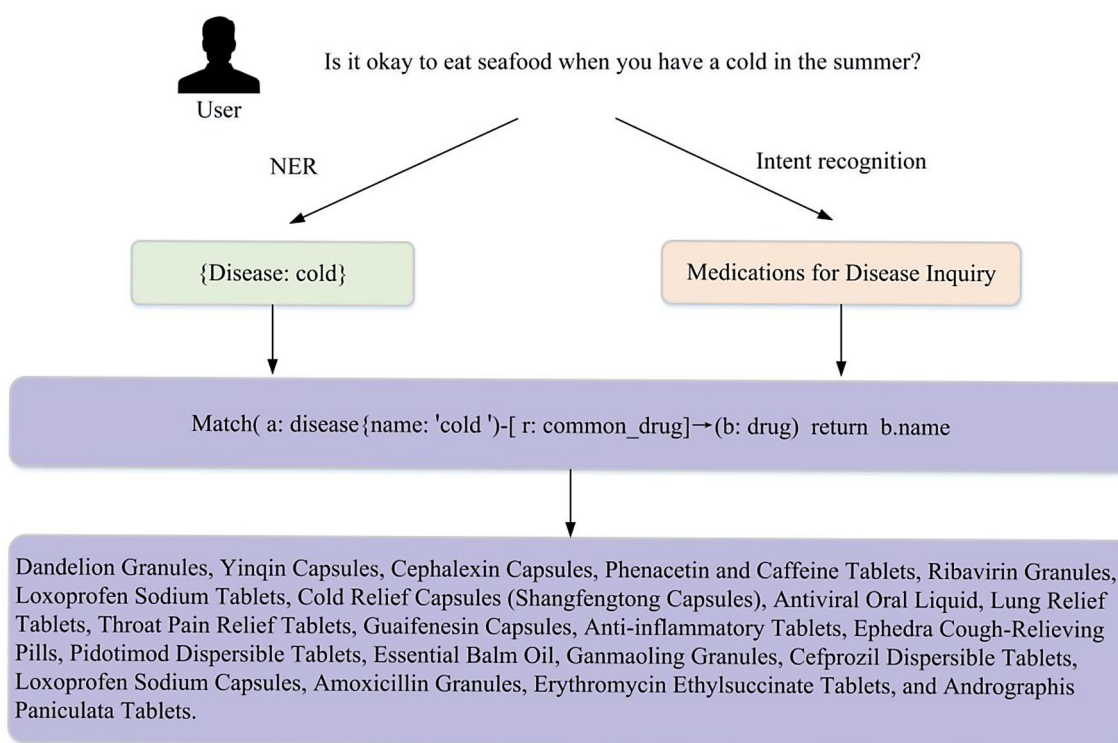


Fig. 5 Prompt engineering

dynamically queried using Cypher query language. NER model integrates BERT model with BiGRU. BERT model encodes contextual information within the sentence, while BiGRU enhances the model's ability to capture sequential dependencies. The model outputs a sequence of labels, with each label indicating the role of a word within the sentence. Regarding model implementation, the system leverages the contextual learning and chain-of-thought capabilities of LLMs. A step-by-step reasoning process decomposes complex questions into multiple sub-questions, gradually constructing the required answers based on prompts derived from the knowledge graph. Moreover, limiting the scope of context usage prevents over-reliance on the model's

pre-existing knowledge, ensuring that all responses are generated exclusively from the provided prompt information.

2.5 Enhancing QA Generation with RAG

This study implements a Chinese medical QA interface using Vue.js for front-end development, providing a streamlined user interaction platform. Upon receiving a medical query through the input interface, the system initiates a structured processing pipeline. First, NER module identifies clinically relevant entities (e.g., disease names, symptoms), while the intent recognition module concurrently categorizes user objectives into predefined classes such as diagnostic

advice, treatment plans, or medication usage. The extracted entities and intent classifications are subsequently routed to the knowledge graph module, which retrieves contextually relevant medical knowledge, including entity attributes and evidence-based guidelines.

The retrieved structured medical data and original user query are integrated into an RAG pipeline, where prompt engineering dynamically combines these elements to guide LLMs in generating responses. To ensure accuracy, the system employs multi-stage validation: treatment recommendations or diagnostic claims generated by LLMs are cross-referenced against the knowledge graph to verify alignment with established medical knowledge. This “retrieve-generate-validate” workflow significantly mitigates hallucination risks while enhancing response credibility.

CMedRAGBot framework supports five state-of-the-art LLMs to accommodate diverse application requirements:

- ChatGPT-4o and ChatGPT-01: developed by OpenAI, are optimized for conversational depth and multi-turn interactions, with the latter fine-tuned for medical specificity in dosage and contraindication descriptions.
- DeepSeek-R1: developed in China, is an open-source domain-specialized model integrating retrieval augmentation, excelling in knowledge-intensive medical QA tasks.
- Llama-3.3-70B-Instruct: developed by Meta, is instruction-tuned for multi-task adaptability, enabling precise medical entity generation through prompt engineering.
- Gemini 2.0 Flash: developed by Google, balances low-latency inference with flexible generation, suitable for real-time clinical interactions.

During response generation, users select their preferred LLM, which processes the knowledge-augmented prompts derived from the fused inputs of retrieved evidence and original queries. The system finalizes outputs through iterative validation, delivering clinically coherent advice structured as “diagnostic guidance – scientific rationale – risk advisories”. By synergizing RAG workflows with multi-model integration, CMedRAGBot demonstrates enhanced performance and practical utility in Chinese medical QA applications.

3 Experiments

3.1 Impact of CMedRAGBot on LLMs

To validate the performance of CMedRAGBot, as of February 25, 2025, five widely recognized state-of-the-art LLMs,

including ChatGPT-4o, ChatGPT-o1, DeepSeek-R1, Llama-3.3-70B-Instruct, and Gemini 2.0 Flash, are selected as benchmarks. The experimental test set is constructed from single-choice and multiple-choice question banks spanning China’s Clinical Medical Qualification Examinations, Residency Standardization Training Examinations, and Postgraduate Entrance Examinations from 2000 to 2023, covering diverse aspects of medical knowledge and question types such as disease identification, treatment recommendations, and medication usage.

In the experiments, a comparative analysis is conducted using identical test questions to evaluate the performance of LLMs both with and without the integration of CMedRAGBot. Given the inherent randomness of LLMs, which can result in slight variations in responses to the same question, several strategies are adopted to ensure that the observed performance differences are attributable to the incorporation of CMedRAGBot. Specifically, the temperature parameter is set to 0.1 to reduce sampling randomness; prompt engineering is optimized to guide the models in generating structured and standardized responses; the output is further constrained using a knowledge graph to ensure response stability. Through these measures, it is demonstrated that the differences in the answers are primarily attributable to the incorporation of CMedRAGBot rather than to the intrinsic randomness of LLMs.

From these question banks, 1,000 items are randomly selected to evaluate the models’ baseline accuracy without CMedRAGBot. The same test set is then used to assess the accuracy gains after integrating CMedRAGBot, as shown in Table 4. Based on the experimental results, a comparative bar chart in Fig. 6 more intuitively illustrates the differences among the various models.

From the experimental results, it has been shown that all models exhibit accuracy improvements upon integration with CMedRAGBot. Specifically, ChatGPT-4o’s accuracy rises from 66.5% to 76.5%, ChatGPT-o1’s from 73.3% to 80.8%, DeepSeek-R1’s from 79.1% to 86.2%, and

Table 4 The impact of CMedRAGBot on LLMs

Model	Size($\times 10^9$)	Accuracy(%)
ChatGPT-4o	-	66.5
ChatGPT-4o CMedRAGBot	-	76.5
ChatGPT-o1	-	73.3
ChatGPT-o1 CMedRAGBot	-	80.8
DeepSeek-R1	70	79.1
DeepSeek-R1 CMedRAGBot	70	86.2
Llama-3.3-70B-Instruct	70	81.8
Llama-3.3-70B-Instruct CMedRAGBot	70	87.4
Gemini 2.0 Flash	-	78.1
Gemini 2.0 Flash CMedRAGBot	-	83.1

Bold represents the accuracy achieved after incorporating CMedRAGBot, with the highest accuracy reaching 87.4%

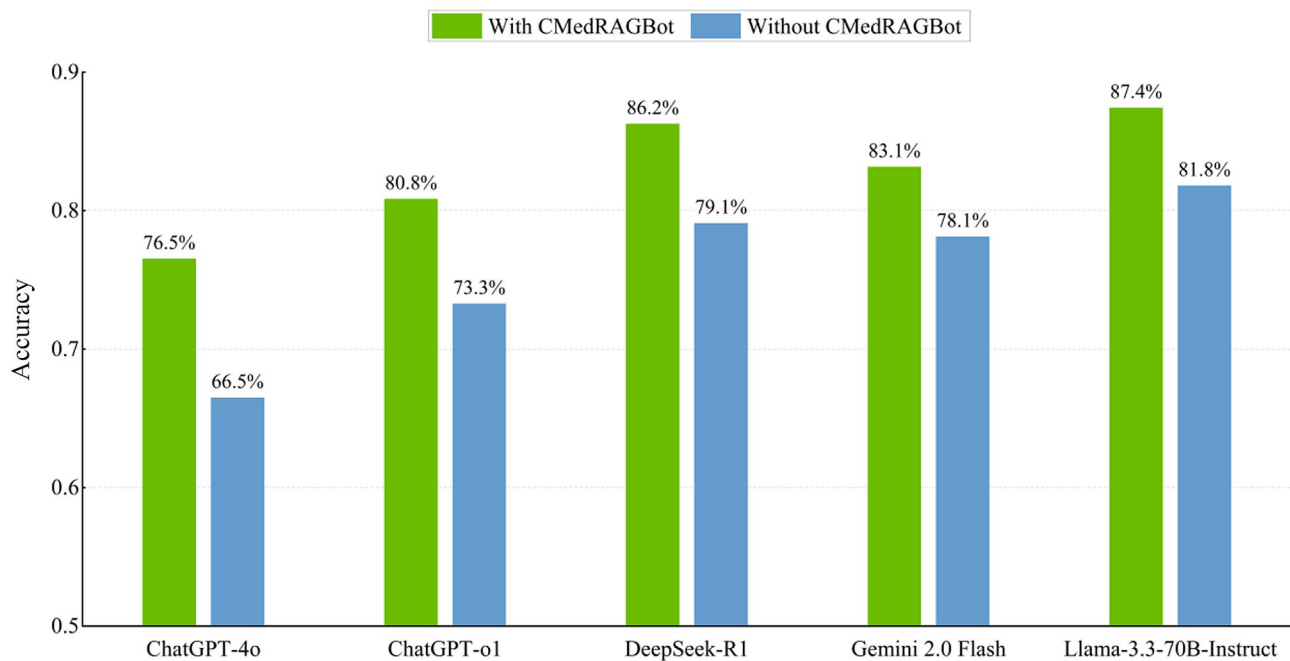


Fig. 6 Comparison of NER performance with and without data augmentation. Blue represents the performance of LLMs without the integration of CMedRAGBot, while green represents their performance following its integration

Llama-3.3-70B-Instruct's from 81.8% to 87.4%. Notably, the two models including 7×10^{10} parameters—DeepSeek-R1 and Llama-3.3-70B-Instruct—achieve accuracies of 86.2% and 87.4%, respectively, after incorporating retrieval augmentation, highlighting the synergistic effect of large model size and retrieval systems in knowledge-intensive tasks. These findings demonstrate that CMedRAGBot's dynamic retrieval of relevant evidence chains from a medical knowledge base significantly boosts performance in complex medical QA. Moreover, it effectively mitigates knowledge hallucinations, underscoring the necessity and efficacy of retrieval augmentation in specialized medical domains.

3.2 Ablation Experiment

In the ablation experiments, different modules are systematically removed from the model to evaluate each component's contribution to overall performance. By comparing the model's performance on medical QA tasks with and without the inclusion of NER and intent recognition modules, the specific roles of these components in enhancing accuracy, generalization, and task efficiency are clearly demonstrated. The primary aim of these ablation experiments is to verify the effectiveness of each component within the proposed model, thereby facilitating further optimization of the model architecture and strengthening its practical applicability in Chinese clinical QA.

3.2.1 NER

In our model, three data augmentation strategies are utilized: entity replacement, entity masking, and entity concatenation. To evaluate the effectiveness of the model in entity recognition, the Chinese pre-trained language model 'Chinese-roberta-wwm-ext' is selected under identical training settings to compare model performance with and without data augmentation, thereby observing the impact of data augmentation strategies. 'Chinese-roberta-wwm-ext' model, released by Harbin Institute of Technology, is a Chinese adaptation of RoBERTa pre-trained with whole word masking, demonstrating strong performance on various Chinese natural language processing tasks.

The experiment utilized the medical NER dataset described in Sect. 2.2.1, which includes a variety of entity types and annotated information. The AdamW optimizer is employed, utilizing the cross-entropy loss function to optimize the sequence labeling task. The training process spans 30 epochs, with dynamic adjustments made based on validation set performance to prevent overfitting, and a batch size of 60 is used. The experimental results are presented in Table 5. Based on the experimental results, a comparative bar chart in Fig. 7 more intuitively illustrates the differences among the various models.

Bold represents the metrics attained post-augmentation.

The results indicate that the data augmentation strategy significantly improves the performance of the 'Chinese-roberta-wwm-ext' model on NER task. Figure 7 illustrates

Table 5 NER performance with and without data augmentation

NER model	Accuracy	Precision	Recall	F1 score
without_data_augmentation	98.84%	94.43%	96.13%	95.27%
with_data_augmentation	99.40%	97.12%	97.98%	97.55%

that the model with data augmentation outperforms the model without augmentation across all evaluation metrics. F1 score reaches 97.55%, representing an approximate 2.28% point improvement compared to the model without data augmentation. These findings indicate that increasing the diversity and robustness of the training data enables the model to better capture subtle differences in entity recognition.

Precision and recall increase to 97.12% and 97.98%, respectively, indicating that the data augmentation strategy effectively reduces false positives, thereby enhancing the model's accuracy in determining entity categories. Overall, the use of various augmentation strategies enhances the model's robustness and generalization capability, enabling it to more accurately and comprehensively recognize and classify entities in medical texts. This significantly improves the model's accuracy and reliability in handling complex medical text data.

3.2.2 Intent Recognition

This section explores the impact of intent recognition on CMedRAGBot model's performance in medical QA tasks. An ablation study was conducted to analyze the impact of

removing or retaining the intent recognition module on overall model performance by establishing a control experiment.

(1) Complete Model: In the complete RAG model, the intent recognition module is retained. In this configuration, the user's medical question first passes through the intent recognition module, which extracts the intent and guides the downstream document retrieval and generation components to find the most relevant answer.

(2) Without Intent Recognition Module: In this configuration, the intent recognition module is omitted. The input question is directly fed into the retrieval and generation components of RAG model without intent recognition."

In the ablation experiment, the removal of the intent recognition module is observed to significantly degrade the model's QA performance. For instance, when the question 'My mother has recently been experiencing chest tightness and palpitations, especially at night when she is trying to rest. Sometimes she has trouble breathing and occasionally coughs up pink, frothy sputum. What could be the cause of this? She has a history of high blood pressure' was input, the impact of intent recognition on the model's response is illustrated in Fig. 8.

In the response without intent recognition, LLM provided only general suggestions, lacking specificity and failing to offer detailed explanations or guidance based on the presented symptoms. Conversely, the response with intent recognition addressed all aspects of the user's concerns, providing more comprehensive information and a more in-depth response. It helps the user understand potential

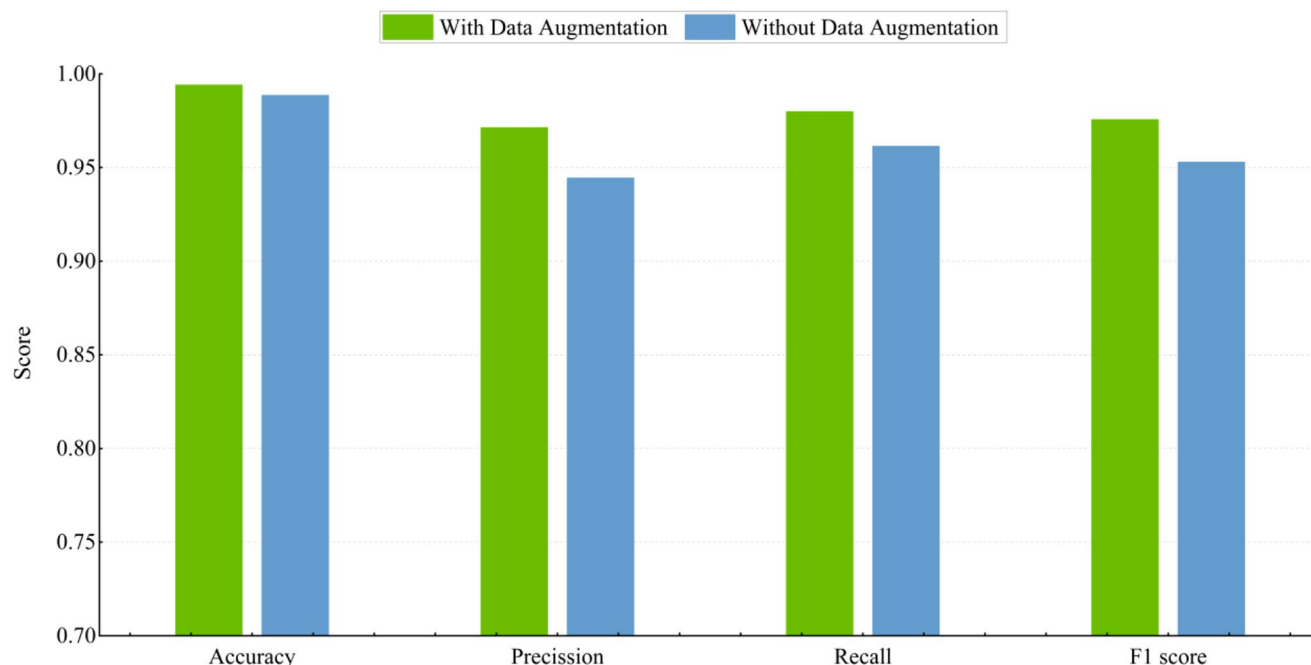


Fig. 7 Comparison of NER performance. Blue represents the performance of NER without data augmentation, while green represents the performance following its integration

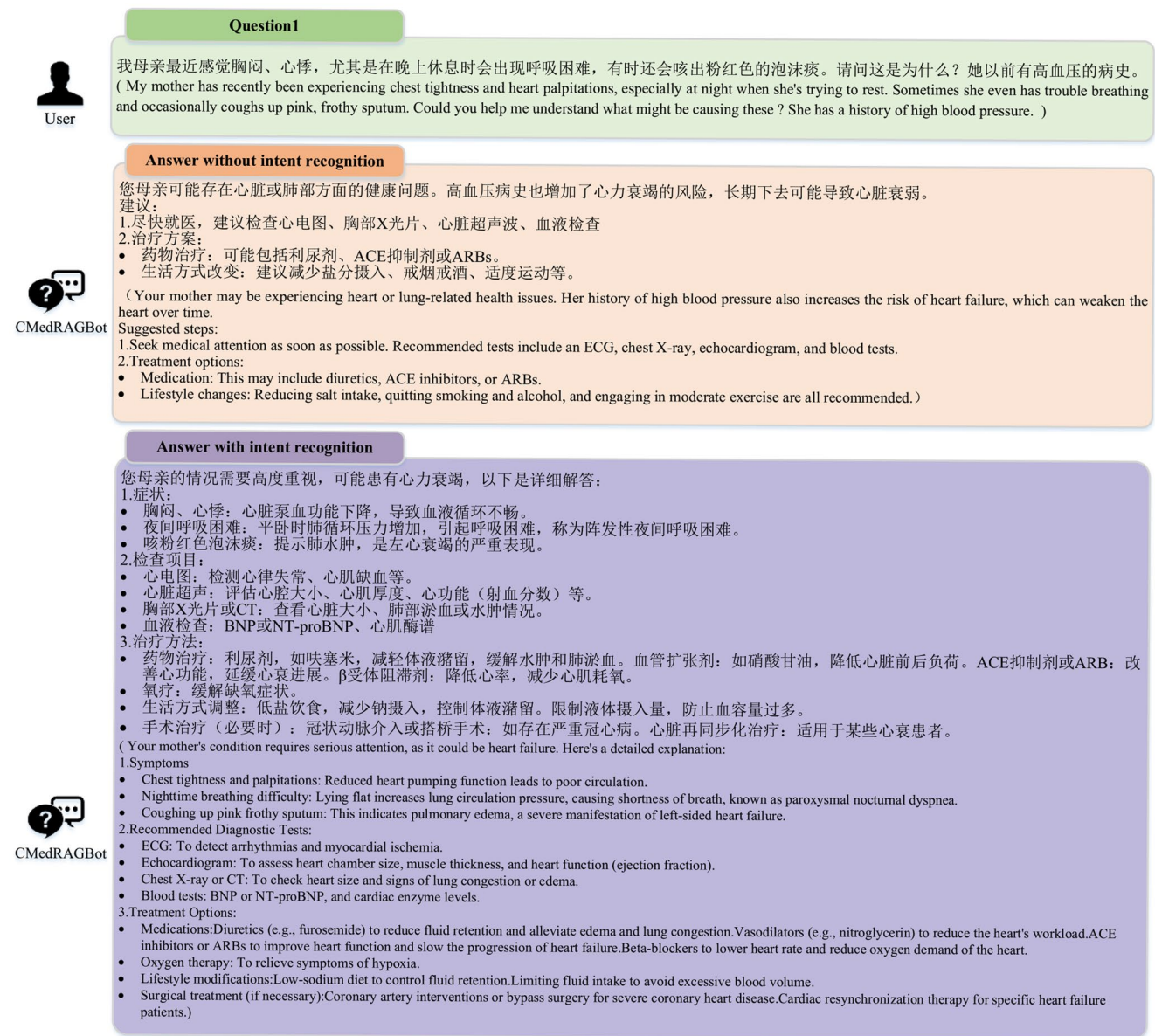


Fig. 8 Comparison of model performance with and without intent recognition

disease types, explains the relationship between high blood pressure and heart failure, and clarifies the underlying cause of the condition. Additionally, the model provided a detailed analysis of the clinical significance of each symptom, emphasizing the urgency of the situation. Furthermore, specific diagnostic tests and their purposes were listed, providing more targeted guidance. The response includes detailed treatment plans for the condition, encompassing medications and lifestyle modifications. Moreover, medical terminology is incorporated, accompanied by layman's explanations to enhance the response's credibility.

The experiments demonstrate the crucial role of the intent recognition module in enhancing model performance. The

intent recognition module enables RAG model to accurately determine the core intent of the user's question, rather than relying solely on keyword matching. Many user queries involve more than simple fact retrieval; they require multi-level reasoning and knowledge integration. By accurately identifying the intent of the query, the model can retrieve more relevant documents, thereby improving the precision of the final response. Moreover, the intent recognition module demonstrates particular advantages when handling long and complex queries. Without this module, the model often becomes entangled in irrelevant information and fails to focus on the critical aspects of the query, thereby reducing response quality.

4 Conclusion and Discussion

This research addresses the challenges of knowledge acquisition and comprehension in Chinese clinical medical QA by introducing CMedRAGBot, a Retrieval-Augmented Generation (RAG) and knowledge graph-based model. Through the integration of NER and intent recognition, CMedRAGBot enhances LLM performance in the medical domain. Specifically, we employ a Chinese-RoBERTa+BiGRU architecture within RAG framework to build and refine NER module. An intent recognition component is further incorporated to precisely classify user requests—such as identifying disease symptoms, treatment options, or recommended medications. Coupled with in-context learning and prompt engineering, this approach substantially improves the model's ability to handle complex queries. Deployed on five LLMs and tested against a professional Chinese medical question bank, CMedRAGBot demonstrates improved accuracy across all models, with gains of up to 10%. Ablation experiments reveal that employing data augmentation boosts NER F1 score from 95.27% to 97.55%, highlighting notable improvements in both recognition accuracy and generalization. Moreover, intent recognition significantly enhances CMedRAGBot's understanding of complex inquiries, thereby increasing answer precision.

Despite these advances, certain limitations remain. For instance, the model lacks self-training capabilities and cannot automatically update its knowledge base, while the current Chinese-RoBERTa+BiGRU-based NER model is tailored for Chinese QA alone. Future work will explore semi-supervised learning and self-training approaches to achieve incremental learning for NER module, along with continual fine-tuning strategies (e.g., LoRA and PEFT) that enable LLMs to optimize their responses over time. Additionally, expanding to multilingual pre-trained models will allow the system to support medical QA in multiple languages and facilitate large-scale testing to further validate its practicality and reliability.

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Data Availability The data in this study has been desensitized and does not involve usability issues.

Declaration

Competing interests The authors declare that they have no known competing financial interests or presinal relationships that could have appeared to influence the work reported in this paper.

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